

Motivation

- Many scientific states are **fields** $u(x, t)$ evolving over space and time, such as velocity, pressure, stress, concentration, water depth, or electromagnetic intensity.
- Their evolution is determined jointly by **geometry, boundary/initial conditions, coefficients, forcing, and interventions**; a fixed observation array is only one discretized view of that system.
- A scientific world model should reuse **one learned latent state for many queries**: forecasting, data assimilation, inverse inference, control, active sensing, and uncertainty-aware decisions.

Central Position

A **PDE-native world model** aligns its representation, prediction targets, constraints, and interventions with field-governed systems. It is broader than a task-specific surrogate because the same learned state should support prediction, inference, and decision queries.

$$\mathcal{W}_\theta : (C, \Omega, \mathcal{B}, \mu, f, a, \mathcal{G}) \mapsto \hat{T}.$$

Here C is partial context data; Ω is the physical domain; $\mathcal{B}$ contains boundary and initial conditions; μ, f , and a denote coefficients, forcing, and an intervention; $\mathcal{G}$ is an optional governing-equation scaffold; and $\hat{T}$ is the requested field, parameter, uncertainty, or decision quantity.

Five Desiderata

- Field-native representation**: encode fields, meshes, graphs, point sensors, and multiresolution states without treating one grid as the physical state itself.
- Operator-valued prediction**: learn maps between function spaces, enabling queries at new coordinates or resolutions rather than only fixed-array regression.
- Geometry & boundary awareness**: condition explicitly on domains and constraints so predictions remain meaningful when shapes, meshes, or operating regimes change.
- Latent physical consistency**: train or audit for conservation, stability, symmetry, calibration, and boundary adherence—not latent similarity alone.
- Intervention readiness**: treat controls, forcing, geometry, and sensing changes as model inputs for counterfactual rollouts and decisions.

Positioning

- Neural operators** learn resolution-aware maps between function spaces, but are commonly trained for one prescribed simulator query.
- Scientific foundation models** emphasize reusable pretraining and transfer, with most evaluations centered on forecasting or solution prediction.
- Perceptual world models / JEPA** motivate prediction in representation space, but their primitive targets are usually image patches, video, or generic sensory features.
- Operator-JEPA** combines these directions: shared latent field prediction with explicit physics checks and action-conditioned, multi-query readout.

Selected References

Assran et al. (2023), *Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture*.
Kovachki et al. (2023), *Neural Operator: Learning Maps Between Function Spaces*.
Li et al. (2021), *Fourier Neural Operator for Parametric Partial Differential Equations*.
Tong and Chen (2026), *From Sparse Sensors to Continuous Fields: STRIDE for Spatiotemporal Reconstruction*.

PDE-Native World Model Architecture

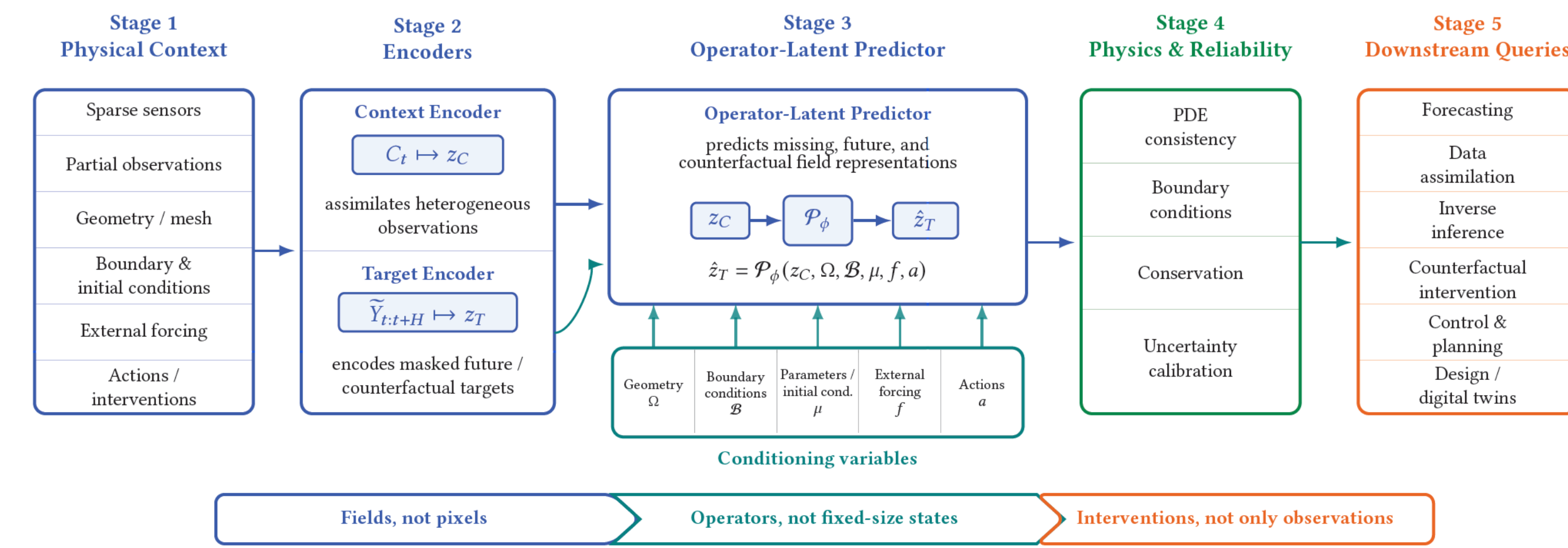


Figure 1. Operator-JEPA with operator-latent prediction, physics constraints, and downstream queries.

Operator-JEPA

1. Context field encoder

$$z_C = E_C(C, \Omega, \mathcal{B}, \mu),$$

maps sparse, partial, noisy, or multiresolution observations C to the latent field state z_C . E_C may process meshes, graphs, point clouds, or sensor histories.

2. Target field encoder

$$z_T = E_T(T),$$

embeds target T as z_T , where T can be a masked region, future field, hidden parameter, counterfactual outcome, or risk field.

3. Operator-latent predictor

$$\hat{z}_T = \mathcal{P}_\phi(z_C, \Omega, \mathcal{B}, \mu, f, a, \mathcal{G}),$$

predicts the target representation under geometry, forcing f , intervention a , and physics scaffold $\mathcal{G}$. ϕ denotes learned predictor parameters.

4. Physics & reliability

$$\mathcal{L} = d(\hat{z}_T, z_T) + \lambda_{\text{PDE}} \mathcal{R}_{\text{PDE}} + \lambda_{\text{BC}} \mathcal{R}_{\text{BC}} + \lambda_{\text{cons}} \mathcal{R}_{\text{cons}},$$

combines latent prediction with PDE-residual, boundary-condition, and conservation terms.

5. Query heads: $Q_\psi(z_C, \hat{z}_T)$ decodes forecasts, hidden parameters, intervention scores, controls, designs, anomalies, or the value of a new measurement.

One Latent Model, Many Scientific Queries

Query and target	Task family
Missing field $z_{\Omega \setminus C}$	Assimilation, super-resolution, sensor fusion
Future state $z_{t+\tau}$	Simulation, forecasting, digital twins
Inverse quantity z_μ	Parameters, source inversion, closure discovery
Counterfactual $z_{t+\tau} \mid a$	Intervention analysis, what-if simulation
Control / planning a^*	Infrastructure, robotics, flow control
Design / sensing Ω^*, x_{new}	Materials, devices, adaptive measurement

Flood Digital-Twin Exemplar

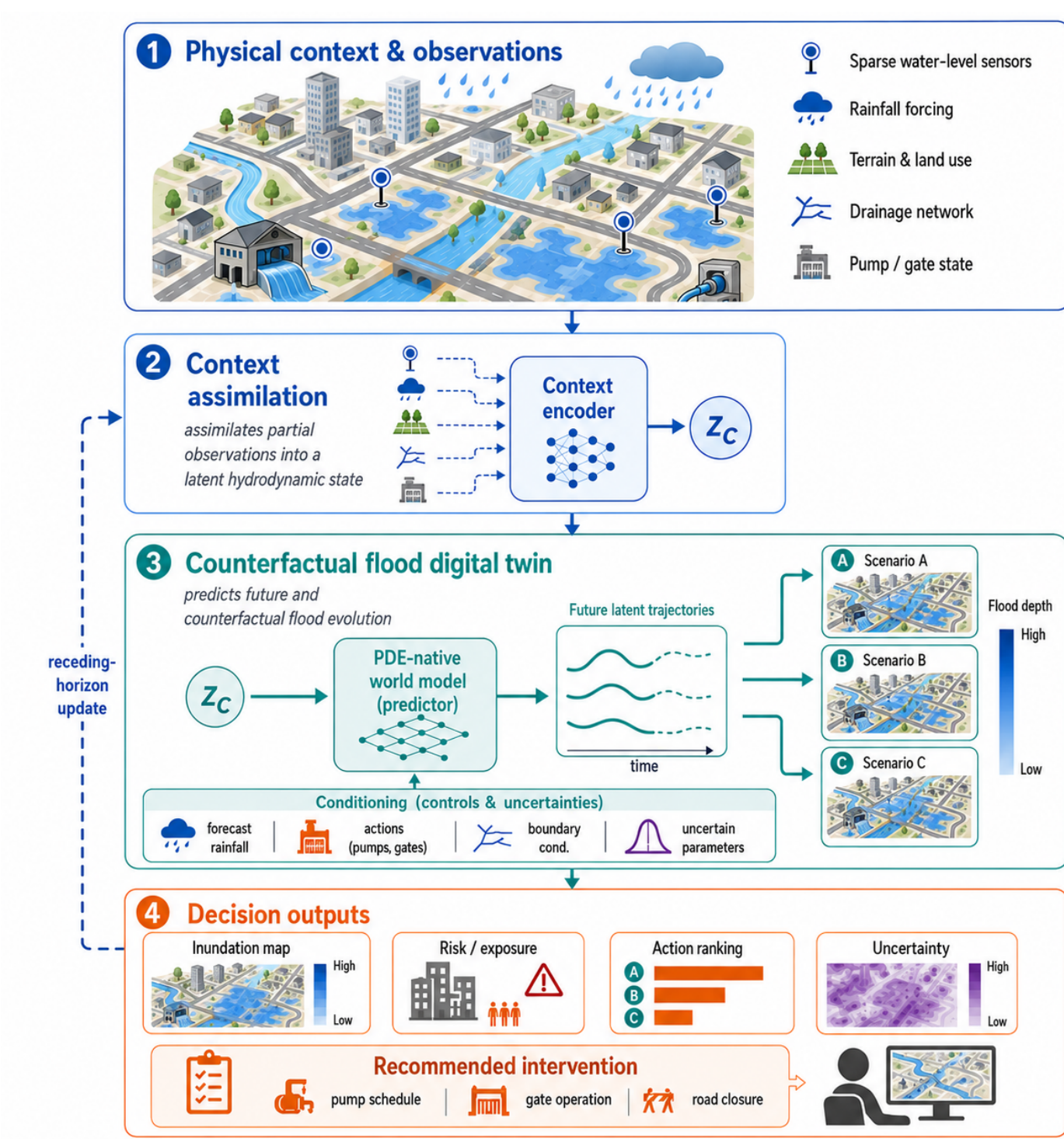


Figure 2. Conceptual flood digital twin as a PDE-native world-model query: assimilate sparse observations, roll out intervention-conditioned futures, and decode inundation, exposure, ranking, and uncertainty.

Counterfactual Flood Intervention

At time t , heterogeneous context is summarized as

$$C_t = \{h(x_i, t), q(x_j, t), r_t(x), \eta(x), g_d, s_t\},$$

with sparse water level/discharge (h, q) , rainfall r_t , terrain η , drainage graph g_d , and infrastructure state s_t . The encoder assimilates C_t into hydrodynamic latent state z_C .

Candidate actions include **pump schedules, gate operations, diversions, or road closures**. A receding-horizon controller ranks them through

$$a^* = \arg \min_{a \in \mathcal{A}} \mathbb{E}_{r_{t:t+H}, \mu} J(D(\mathcal{P}_\phi(z_C, r_{t:t+H}, a, \Omega, \mathcal{B}, \mu))),$$

Here $\mathcal{A}$ is the feasible action set, H the planning horizon, and the expectation covers uncertain rainfall $r_{t:t+H}$ and parameters μ . Decoder D produces inundation/velocity/risk maps. J combines exposure, hydraulic constraints, intervention cost, and uncertainty.

Research Agenda

- Scientific tokenization**: represent fields, operators, geometries, meshes, sensors, parameters, and boundaries while retaining physical scale and units.
- Geometry & discretization transfer**: disentangle an underlying field from its grid, mesh, point-cloud, graph, manifold, or adaptive-discretization view.
- Trustworthy latent physics**: audit conservation, boundary adherence, rollout stability, plausibility, and calibration, including under distribution shift.
- Intervention-valid learning**: combine simulations, experiments, causal structure, and domain theory to test predicted action responses.
- Evaluation beyond pointwise error**: report transfer, intervention ranking, decision regret, calibration, stability, and scientific usefulness.